OPEN INSTITUTE OF COMPUTER SCIENCE & ENGINEERING

चिन्तनात् नवसृष्टिः

MATHEMATICS • ALGORITHMS • ARTIFICIAL INTELLIGENCE

OICSE

CS101 Programming with Modern C++

Purpose

To establish the programming foundation required for Data Structures and Algorithms (DSA), Competitive Programming (CP), AI backend development, and general software engineering through the study of C++ syntax, memory management, Object-Oriented Programming (OOP), and the Standard Template Library (STL).

Syllabus

Week 1: Introduction to C++, Program Structure, Compilation Process, Variables, Fundamental Data Types, and Basic Input/Output.

Week 2: Control Flow Statements, Loop Constructs, Function Definitions, Parameter Passing, Return Values, and Scope Resolution.

Week 3: Debugging Methodologies, Compiler Diagnostics, Assertions, Constant Expressions, String Manipulation, and Fundamental Type Properties.

Week 4: Arithmetic and Logical Operators, Bitwise Operations, Operator Precedence, Type Conversions, Namespaces, and Header Organization.

Week 5: References, Pointers, Dynamic Memory Concepts, Structures, Enumerations, and Const Correctness.

Week 6: Class Definitions, Object Instantiation, Constructors, Destructors, Encapsulation, Member Functions, and Copy Semantics.

Week 7: Advanced Class Design, Static Members, Friend Functions, Operator Overloading, Inheritance, Polymorphism, and Object Relationships.

Week 8: Standard Template Library Philosophy, Iterators, Sequence Containers, Associative Containers, and Standard Algorithms.

Week 9: Function Overloading, Default Arguments, Function Templates, Generic Programming Principles, and Type Deduction.

Week 10: Stream Handling, I/O Formatting, File Processing, String Streams, and Optimized Input/Output Techniques.

Week 11: Move Semantics, Resource Ownership, Smart Pointers, and Modern Memory Management Strategies.

Week 12: Integrated Problem Solving, Algorithmic Implementation Patterns, Competitive Programming Strategies, and Final Project Development.

Resources

- LearnCpp.com
- NPTEL – Programming in Modern C++ (IIT KGP)

CS102 Mathematical Foundations for Machine Learning

Purpose

To establish the mathematical foundation required for Machine Learning, Artificial Intelligence, Data Science, and optimization-based algorithms through the study of Linear Algebra, Calculus, Probability, and Mathematical Optimization. The course develops the theoretical understanding and practical implementation skills required for ML algorithms, including dimensionality reduction, regression, classification, and Support Vector Machines.

Syllabus

Week 1: Vectors in Machine Learning, Basics of Matrix Algebra, Vector Spaces, Subspaces, Basis, and Dimension.

Week 2: Linear Transformations, Norms and Vector Spaces, Orthogonal Complements, Projection Mapping, Eigenvalues and Eigenvectors, Special Matrices and Their Properties.

Week 3: Spectral Decomposition, Singular Value Decomposition (SVD), Low-Rank Approximations, Python Implementation of SVD and Low-Rank Approximation.

Week 4: Principal Component Analysis (PCA), Python Implementation of PCA, Linear Discriminant Analysis (LDA), Python Implementation of LDA.

Week 5: Least Square Approximation, Minimum Norm Solutions, Linear Regression, Multiple Regression, and Logistic Regression.

Week 6: Classification Metrics, Gram-Schmidt Process, Polar Decomposition, Minimal Polynomial, Jordan Canonical Form, and Applications of Matrix Methods in Machine Learning.

Week 7: Fundamentals of Calculus, Gradients, Jacobians, Chain Rule, and Change of Variables.

Week 8: Calculus Implementation in Python, Convex Sets, Convex Functions, Properties of Convex Functions, and Introduction to Optimization.

Week 9: Numerical Optimization in Machine Learning, Gradient Descent, and Advanced Optimization Algorithms Used in Machine Learning.

Week 10: Optimization Using Python, Probability Review, Bayes' Theorem, Random Variables, Expectation, and Variance.

Week 11: Discrete and Continuous Probability Distributions, Joint Probability, Covariance, Introduction to Support Vector Machines (SVM), and Error Minimization as Linear Programming Problems.

Week 12: Lagrangian Multipliers, Concepts of Duality, Hard and Soft Margin Classification, Support Vector Machines Implementation in Python.

Resources

- NPTEL – Essential Mathematics for Machine Learning (IIT Roorkee)
- *Mathematics for Machine Learning* — Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong

PE001 Programming with Python

Purpose

To establish a strong foundation in Python programming required for Artificial Intelligence, Machine Learning, Data Science, automation, and software development through the study of Python syntax, procedural programming, control flow, functions, exception handling, file processing, object-oriented programming, regular expressions, libraries, and software testing.

Syllabus

Week 1: Functions, Variables, Basic Input and Output, Expressions, and Python Program Structure.

Week 2: Conditional Statements, Boolean Logic, Relational Operators, and Decision Making.

Week 3: Loop Constructs, Iteration, Nested Loops, and Basic Algorithmic Problem Solving.

Week 4: Exception Handling, Error Management, Assertions, and Defensive Programming.

Week 5: Python Libraries, Modules, Packages, Command-Line Arguments, and Code Reusability.

Week 6: Unit Testing, Test-Driven Development Fundamentals, Assertions, and Code Validation.

Week 7: File Input/Output, Reading and Writing Files, CSV Processing, and Persistent Data Storage.

Week 8: Regular Expressions, Pattern Matching, Text Processing, and String Manipulation.

Week 9: Object-Oriented Programming, Classes, Objects, Attributes, Methods, Constructors, and Encapsulation.

Week 10: Advanced Python Topics, Iterators, Comprehensions, Pythonic Programming Practices, and Standard Library Utilities.

Week 11: Integrated Problem Solving, Modular Program Design, Code Organization, and Software Development Best Practices.

Week 12: Final Project Development, Testing, Documentation, and Project Presentation.

Resources

- CS50's Introduction to Programming with Python (Harvard University, edX)
- Python Documentation

IE001 Foundations of Entrepreneurship

Purpose

To develop the entrepreneurial mindset, innovation skills, and business fundamentals required to identify opportunities, create customer value, develop viable products, formulate business models, understand startup finance, and effectively communicate and pitch entrepreneurial ventures.

Syllabus

Week 1: Introduction to Innovation and Entrepreneurship, Creativity, Entrepreneurial Mindset, India's Startup Ecosystem, and Role of Entrepreneurs.

Week 2: Opportunity Identification, Idea Generation, Problem Discovery, Emerging Technology Opportunities, Artificial Intelligence, and Healthcare Entrepreneurship.

Week 3: Customer Value Proposition, Market Segmentation, Customer Discovery, Business Models, and Value Creation.

Week 4–5: New Product Development, Design Thinking, Minimum Viable Product (MVP), Product Development Lifecycle, Case Studies, and Intellectual Property Fundamentals.

Week 6–7: Marketing Fundamentals, Go-to-Market (GTM) Strategy, Branding, Product Differentiation, Pricing Strategies, Customer Journey, Digital Marketing, Sales Strategies, B2B Marketing, and Sales Channels.

Week 8–9: Startup Finance, Accounting Fundamentals, Financial Statements, Unit Economics, Product Costing, Business Planning, Financial Planning, Fundraising, Venture Capital, Incubation, Grants, and Investment Strategy.

Week 10–11: Business Strategy, Leadership, Business Models, Business Model Canvas, Strategic Planning, Social Entrepreneurship, and Entrepreneurial Leadership.

Week 12: Communication Skills, Value Creation, Investor Pitching, Startup Presentations, Founder Insights, and Course Integration.

Resources

- NPTEL – Foundations of Entrepreneurship (IIT Bombay)
- *The Lean Startup* — Eric Ries

Project Part – 1

Objective

Develop an Inventory & Point of Sale (POS) Management System that applies object-oriented programming principles, modular software design, and efficient data management techniques to build a complete command-line application for managing inventory, sales, purchases, suppliers, customers, and business reports.

Technologies

- Modern C++17
- Python 3

Skills Learned

- Object-Oriented Programming (OOP)
- Modular Software Design
- Standard Template Library (STL)
- File Handling and Data Persistence
- Searching and Sorting Algorithms
- Exception Handling and Data Validation
- Modern Memory Management (Smart Pointers and RAII)
- Command-Line Application Development
- Version Control using Git
- Basic Business Logic and Inventory Management

Expected Deliverables

A modular command-line Inventory & Point of Sale (POS) Management System capable of maintaining product inventories, managing suppliers and customers, recording purchases and sales, automatically updating stock levels, generating invoices and business reports, calculating inventory valuation and profits, and persisting all data using file storage while adhering to modern C++ software engineering practices.